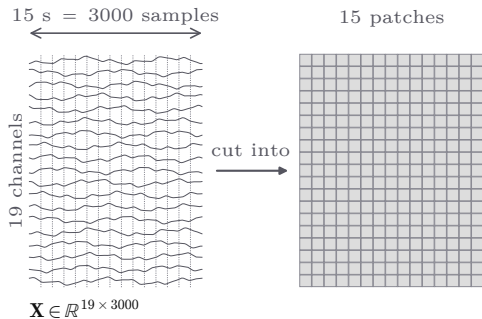


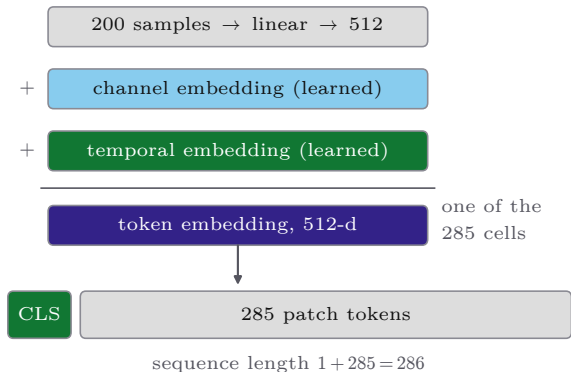
(a) From window to token grid



Each cell is one channel over one second: 200 samples.
 $N = C \times P = 19 \times 15 = 285$ tokens (Eq. 4.7).

Cutting along both axes at once is what lets a token be withheld in time, in space, or in both.

(b) From token grid to sequence



The CLS token is learned, holds no patch content and is never masked. Its final encoder state is the window representation every downstream head reads.

Each proposed variant changes one piece of this:
BrainLM-EEG-RoPE replaces the channel embedding,
BrainLM-EEG-Microstate adds a decoder target.